– Task 2: Build a Simple Predictive Model

Objective

Use a dataset to build a basic predictive model using **Scikit-learn**. The model should predict one target column based on other features in the dataset.

Dataset

Use the **Titanic Dataset** (commonly used for beginners in Data Science):

Dataset Link:
https://www.kaggle.com/datasets/yasserh/titanic-dataset

(Interns can download the CSV file and use it locally.)

Task Requirements

- Load and explore the dataset
- Preprocess the data:
  - Handle missing values
  - Encode categorical features if required
- Split the dataset into **training and testing sets**
- Train a predictive model (e.g., Linear Regression, Logistic Regression, Decision Tree, etc.)
- Evaluate the model using appropriate metrics such as:
  - Accuracy
  - Mean Squared Error (MSE)
  - R^2 Score (as applicable)
- Clearly document each step and explain the model's results and performance

Deliverables

- **Python script or Jupyter Notebook** containing:
  - Data preprocessing
  - Model training
  - Evaluation
- **Short report** describing:
  - Approach and steps followed
  - Model performance
  - Observations and conclusions